

Smart Home AIoT – Test Data Collection Sheet – Phase C – Mode Change & Cascading

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Instructions for use:

1. Run each test 10 times consecutively (or across sessions, as practical)
2. Record: Date, Time, exact Input given, Actual Output observed, Response Time (use stopwatch), and Pass/Fail
3. Pass criteria: Actual matches Expected description
4. After 10 trials, calculate: Pass Rate %, Average Response Time, Standard Deviation
5. Use Comments for any anomalies or special observations

Statistical formulas:

- Pass Rate = $(\text{Pass Count} / 10) \times 100\%$
- Avg RT = $(\text{response times}) / 10$
- Variance = $(x_i -)^2 / n$
- Std Dev = $\sqrt{\text{variance}}$

Coverage Matrix

Phase	Description	Count
Phase C	Phase C – Mode Change & Cascading	10
	TOTAL	10

Phase C – Mode Change & Cascading

C01	House Mode	Home	Phase C				
Method: Set input_select.house_mode = Home							
Expected: Mode Scene fires + adaptive ON + GPS welcome enabled							
#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
Pass Count: ____/10		Pass Rate: ____%		Avg RT: ____ ms		Std Dev: ____ ms	
Comments:							

Method: Set input_select.house_mode = Away

Expected: 7n switch OFF + lock + curtain close + LINE notify

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
Pass Count: ____/10			Pass Rate: ____%	Avg RT: ____ ms		Std Dev: ____ ms	
Comments:							

Method: Set input_select.house_mode = Security

Expected: Doorbell + monitor active (door open = intrusion)

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ____/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: Set input_select.comfort_mode = Normal

Expected: adaptive light ON, sensors active

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
Pass Count: ____/10			Pass Rate: ____%	Avg RT: ____ ms		Std Dev: ____ ms	
Comments:							

C05

Comfort Mode Sleep

Phase C

Method: Set comfort_mode = Sleep

Expected: ปิดไฟทุกห้อง + curtain close + path-light enabled

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ____/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: Set comfort_mode = Relax

Expected: Tuya warm 60% + auto-fan if >28°C

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
Pass Count: ___/10			Pass Rate: ____%	Avg RT: ____ ms		Std Dev: ____ ms	
Comments:							

C07

Comfort Mode Warm

Phase C

Method: Set comfort_mode = Warm

Expected: Tuya 2700K + ปิดพัดลม

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: เปลี่ยน mode รั้วๆ 3 ครั้ง

Expected: All devices end at correct state

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
Pass Count: ____/10			Pass Rate: ____%	Avg RT: ____ ms		Std Dev: ____ ms	
Comments:							

C09

Concurrent: Home + Sleep

Phase C

Method: Set both

Expected: Path-light ON, adaptive OFF

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							

Pass Count: ___/10

Pass Rate: ____%

Avg RT: ____ ms

Std Dev: ____ ms

Comments:

Method: Change Sleep Normal

Expected: Auto-fan stops, adaptive resumes

#	Date	Time	Input/Command	Actual Output	RT (ms)	P/F	Notes
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
Pass Count: ____/10			Pass Rate: ____%	Avg RT: ____ ms		Std Dev: ____ ms	
Comments:							